CSE402: Optimisation

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1 Lecture 01

Mathematical language of this course is linear algebra (of real vector spaces) and multivariable calculus. In this lecture, we will give a brief review of linear algebra.

§1.1 Review on linear algebra

Notation. Let $a_1, a_2, \ldots, a_n$ be a finite sequence of real numbers. We will denote the sum

$$S = a_1 + a_2 + \cdots + a_n \tag{1.1}$$

by

$$S = \sum_{k=1}^{n} a_k. \tag{1.2}$$

We have the following useful identities:

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \tag{1.3}$$

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}. \tag{1.4}$$

§1.1.1 Vectors

Definition 1.1.1 ($\mathbb{R}$-vector space)**.** Let V be a non-empty set. The set V equipped with two binary operations

$$+ : V \times V \to V$$

called *vector addition*, and

$$\cdot : \mathbb{R} \times V \to V$$

called *scalar multiplication* is called a **vector space over** $\mathbb{R}$ or $\mathbb{R}$**-vector space** if for all $u, v, w \in V$ and all $a, b \in \mathbb{F}$, the following axioms hold:

1. **Closure under addition:**
$$u + v \in V \tag{1.5}$$
2. **Commutativity of addition:**
$$u + v = v + u \tag{1.6}$$
3. **Associativity of addition:**
$$(u + v) + w = u + (v + w) \tag{1.7}$$

4. **Existence of zero vector:** There exists a vector $0 \in V$ such that

$$u + 0 = u \tag{1.8}$$

5. **Existence of additive inverse:** For every $u \in V$, there exists $-u \in V$ such that

$$u + (-u) = 0 \tag{1.9}$$

6. **Closure under scalar multiplication:**

$$au \in V \tag{1.10}$$

7. **Distributivity of scalar multiplication over vector addition:**

$$a(u + v) = au + av \tag{1.11}$$

8. **Distributivity of scalar multiplication over field addition:**

$$(a + b)u = au + bu \tag{1.12}$$

9. **Associativity of scalar multiplication:**

$$a(bu) = (ab)u \tag{1.13}$$

10. **Identity element of scalar multiplication:**

$$1u = u \tag{1.14}$$

Hence, $(V, +, \cdot)$ is called a vector space over the field $\mathbb{R}$. Elements of V are called **vectors** and elements of $\mathbb{R}$ are called **scalars**.

Convention. From now on, vector space means $\mathbb{R}$-vector space, unless stated otherwise.

In this course, we will work with only the following special example of $\mathbb{R}$-vector space.

Example 1.1.2

Let

$$V \coloneqq \mathbb{R}^n = \{[x_1, x_2 \ldots, x_n]^T \mid x_1, x_2, \ldots, x_n \in \mathbb{R}\}. \tag{1.15}$$

For $[x_1, x_2 \ldots, x_n]^T, [y_1, y_2 \ldots, y_n]^T \in V$ and $\alpha \in \mathbb{R}$ define addition $(+)$ and scalar multiplication $(\cdot)$, respectively as follows:

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}, \tag{1.16}$$

$$\alpha \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{bmatrix}. \tag{1.17}$$

Then it is an easy check that $V = \mathbb{R}^n$ forms an $\mathbb{R}$-vector space.

§1.1.2 Linear transformations

In this course, we are mostly concerned with following special functions between vector spaces:

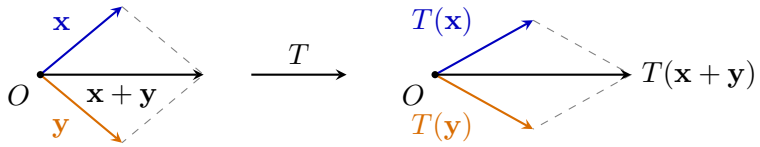
Not sure about that

Definition 1.1.3 (Linear transformation)**.** Let V, W be two vector space. A **linear transformation** from V to W is a function $T : V \to W$ such that for all $\mathbf{x}, \mathbf{y} \in V$ and $\alpha \in \mathbb{R}$ the following holds true:

- **Additivity.**
$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}). \tag{1.18}$$
- **Homogeneity.**
$$T(\alpha \cdot \mathbf{x}) = \alpha \cdot T(\mathbf{x}). \tag{1.19}$$

Additivity and homogeneity property of linear transformation can be visualised as follow:

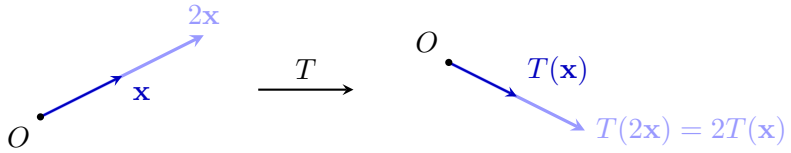
Additivity. The diagonal of the parallelogram represents $\mathbf{x} + \mathbf{y}$. After applying T, the transformed vectors form a parallelogram whose diagonal is $T(\mathbf{x} + \mathbf{y})$.



Thus,

$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}).$$

Homogeneity. For example, doubling $\mathbf{x}$ doubles its image under T.



More generally,

$$T(\alpha\mathbf{x}) = \alpha T(\mathbf{x}) \qquad \text{for every } \alpha \in \mathbb{R}.$$

Combining additivity and homogeneity gives

$$T(a\mathbf{x} + b\mathbf{y}) = aT(\mathbf{x}) + bT(\mathbf{y}).$$

Example 1.1.4

Let $V := \mathbb{R}^n, W := \mathbb{R}^m$. An $m \times n$ matrix M then defines a linear transformation $M : \mathbb{R}^n \to \mathbb{R}^m$ [a] as follows:

For $\mathbf{x} \in \mathbb{R}^n$,

$$M(\mathbf{x}) := M\mathbf{x}. \tag{1.20}$$

Here, M is an $m \times n$ matrix, $\mathbf{x}$ can is an $n \times 1$ matrix. So $M\mathbf{x}$ is an $m \times 1$ matrix. Therefore, M is indeed a function from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Furthermore, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$, the following holds by the properties of

matrix multiplication:

$$M(\mathbf{x}+\mathbf{y}) = M\mathbf{x} + M\mathbf{y}, \tag{1.21}$$

$$M(\alpha\mathbf{x}) = \alpha M\mathbf{x} \tag{1.22}$$

Hence, M is indeed a linear transformation.

[a]Here, we are denoting the linear transformation determined by the matrix M by the same letter M using abuse of notation.

Surprising fact is that an arbitrary linear transformation also comes from a matrix. Before that, we need the idea of basis.

§1.1.3 Basis and dimension of $\mathbb{R}^n$

Consider the vector space

$$\mathbb{R}^2 = \left\{[x,y]^T : x,y \in \mathbb{R}\right\}. \tag{1.23}$$

A natural question is: how many vectors are required to generate every vector in $\mathbb{R}^2$? Answer to this question is particularly useful in the context of linear transformations.

Suppose

$$L : \mathbb{R}^2 \to V$$

is a linear transformation. Because $\mathbb{R}^2$ contains infinitely many vectors, describing which vector maps to what under L would require an infinite amount of information.

However, suppose every vector of $\mathbb{R}^2$ can be generated by a finite collection of vectors

$$v_1, \ldots, v_n \tag{1.24}$$

by taking "linear combinations". Then, instead of knowing the value of L on every vector, we only need to know the values

$$L(v_1), \ldots, L(v_n). \tag{1.25}$$

Indeed, if

$$v = a_1 v_1 + \cdots + a_n v_n, \tag{1.26}$$

then by the linearity of L,

$$L(v) = L(a_1 v_1 + \cdots + a_n v_n) = a_1 L(v_1) + \cdots + a_n L(v_n). \tag{1.27}$$

Therefore, knowing the images of $v_1, \ldots, v_n$ completely determines the linear transformation L on the whole space.

This motivates the following question: What is the smallest collection of vectors required to generate all vectors of a vector space?

The answer leads us to the concepts of spanning sets, linear independence, basis, and dimension:

Definition 1.1.5 (Linear Independence)**.** A collection of vectors

$$\{v_1, v_2, \ldots, v_k\} \tag{1.28}$$

in a vector space V is called *linearly independent* if

$$a_1v_1 + a_2v_2 + \cdots + a_kv_k = 0 \tag{1.29}$$

implies

$$a_1 = a_2 = \cdots = a_k = 0. \tag{1.30}$$

If there exists a non-zero solution of the above equation, then the vectors are called *linearly dependent*.

Intuitively, if a list of vectors are linearly dependent, then at least one of them can be generated by rest of them.

Example 1.1.6

The vectors

$$\begin{bmatrix}1\\1\end{bmatrix}, \qquad \begin{bmatrix}1\\-1\end{bmatrix} \tag{1.31}$$

are linearly independent.

However,

$$\begin{bmatrix}1\\2\end{bmatrix}, \qquad \begin{bmatrix}2\\4\end{bmatrix} \tag{1.32}$$

are linearly dependent because the second vector is a scalar multiple of the first.

Definition 1.1.7 (Span)**.** Let V be a vector space and let

$$S = \{v_1, v_2, \ldots, v_n\} \tag{1.33}$$

be a collection of vectors in V. The span of S, denoted by

$$\text{span}\{v_1, v_2, \ldots, v_n\}, \tag{1.34}$$

is the set of all possible linear combinations of the vectors in S. That is,

$$\text{span}\{v_1, v_2, \ldots, v_n\} = \{a_1v_1 + a_2v_2 + \cdots + a_nv_n : a_1, \ldots, a_n \in \mathbb{R}\}. \tag{1.35}$$

If

$$\text{span}\{v_1, v_2, \ldots, v_n\} = V, \tag{1.36}$$

then we say that $v_1, v_2, \ldots, v_n$ span the vector space V.

A basis is a minimal spanning set, that is, it contains enough vectors to generate the whole space, but no redundant vectors.

Example 1.1.8

Consider the vectors in $\mathbb{R}^2$

$$v_1 = \begin{bmatrix}1\\1\end{bmatrix}, \qquad v_2 = \begin{bmatrix}1\\-1\end{bmatrix}. \tag{1.37}$$

The span of these two vectors is

$$\operatorname{span}\{v_1, v_2\} = \left\{ a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix} : a, b \in \mathbb{R} \right\}. \tag{1.38}$$

Now,

$$a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} a+b \\ a-b \end{bmatrix}. \tag{1.39}$$

Given any vector

$$\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2, \tag{1.40}$$

choosing

$$a = \frac{x+y}{2}, \qquad b = \frac{x-y}{2} \tag{1.41}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \tag{1.42}$$

Therefore,

$$\operatorname{span}\{v_1, v_2\} = \mathbb{R}^2. \tag{1.43}$$

Hence, the vectors v_1, v_2 span $\mathbb{R}^2$.

Definition 1.1.9 (Basis)**.** Let V be a vector space. A set of vectors

$$\mathcal{B} = \{v_1, v_2, \ldots, v_k\} \tag{1.44}$$

is called a *basis* of V if:

1. $v_1, v_2, \ldots, v_k$ span V, i.e.

$$V = \operatorname{span}\{v_1, v_2, \ldots, v_k\}, \tag{1.45}$$

2. $v_1, v_2, \ldots, v_k$ are linearly independent.

Example 1.1.10

Consider the set of vectors

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}. \tag{1.46}$$

We claim that $\mathcal{B}$ is a basis of $\mathbb{R}^2$.
First, let

$$\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \tag{1.47}$$

be an arbitrary vector. We want to express it as a linear combination of the

vectors in $\mathcal{B}$. Suppose

$$\begin{bmatrix} x \\ y \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \tag{1.48}$$

Then,

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a+b \\ a-b \end{bmatrix}. \tag{1.49}$$

Solving the equations

$$a+b=x, \qquad a-b=y, \tag{1.50}$$

we get

$$a = \frac{x+y}{2}, \qquad b = \frac{x-y}{2}. \tag{1.51}$$

Therefore, every vector in $\mathbb{R}^2$ can be written as a linear combination of the vectors in $\mathcal{B}$. Hence,

$$\operatorname{span}(\mathcal{B}) = \mathbb{R}^2. \tag{1.52}$$

Also, the two vectors in $\mathcal{B}$ are linearly independent. Therefore, $\mathcal{B}$ is a basis of $\mathbb{R}^2$. Moreover, the representation

$$\begin{bmatrix} x \\ y \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix} \tag{1.53}$$

is unique. The scalars a and b are called the coordinates of the vector with respect to the basis $\mathcal{B}$.

§1.1.4 Dimension

The number of vectors contained in a basis gives the size of a vector space.

Theorem 1.1.11

Every basis of a finite-dimensional vector space contains the same number of vectors.

Proof. This proof was not shown in the class. Interested students may refer to [Axl24, p. 44]. □

is it examinable?

Definition 1.1.12 (Dimension)**.** The number of vectors in a basis of a vector space V is called the *dimension* of V, denoted by

$$\dim(V).$$

Example 1.1.13

Since every basis of $\mathbb{R}^2$ contains exactly two vectors,

$$\dim(\mathbb{R}^2) = 2.$$

§1.1.5 Basis and dimension of $\mathbb{R}^n$

The standard basis of $\mathbb{R}^n$ is

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \ldots, \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}. \tag{1.54}$$

Every vector

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \tag{1.55}$$

can be written as

$$x_1 e_1 + x_2 e_2 + \cdots + x_n e_n. \tag{1.56}$$

Therefore,

$$\{e_1, e_2, \ldots, e_n\} \tag{1.57}$$

is a basis of $\mathbb{R}^n$.

Theorem 1.1.14

The dimension of $\mathbb{R}^n$ is

$$\dim(\mathbb{R}^n) = n. \tag{1.58}$$

§1.1.6 Eigenvectors and eigenvalues

Suppose we have a linear transformation $L : \mathbb{R}^n \to \mathbb{R}^m$. We are interested in finding a non-zero vector $\mathbf{x} \in \mathbb{R}^n$ such that

$$L\mathbf{x} = \lambda \mathbf{x}, \tag{1.59}$$

where $\lambda \in \mathbb{R}$.

Definition 1.1.15 (Eigenvector and eigenvalue)**.** $\mathbf{x}, \lambda$ (if exists) are called an **eigenvector** and **eigenvalue** for L. The $(\mathbf{x}, \lambda)$ is called an **eigenpair** for L.

Theorem 1.1.16

Let L be a linear transformation. Then $(\mathbf{x}, \lambda)$ is an eigen pair of L if and only if $\det(L - \lambda I) = 0$.

Proof. Suppose λ is an eigenvalue of L. Then there exists a non-zero vector $\mathbf{x} \in \mathbb{R}^n$ such that

$$L\mathbf{x} = \lambda \mathbf{x}. \tag{1.60}$$

Since $I\mathbf{x} = \mathbf{x}$, we have

$$L\mathbf{x} - \lambda I\mathbf{x} = 0, \tag{1.61}$$

or equivalently,

$$(L - \lambda I)\mathbf{x} = 0. \tag{1.62}$$

Since $\mathbf{x} \neq 0$, this homogeneous system has a non-trivial solution. Therefore, $L - \lambda I$ is singular. Hence,

$$\det(L - \lambda I) = 0. \tag{1.63}$$

Conversely, suppose $\det(L-\lambda I) = 0$. Then $L-\lambda I$ is singular, so there exists a non-zero vector $\mathbf{x} \in \mathbb{R}^n$ satisfying

$$(L - \lambda I)\mathbf{x} = 0. \tag{1.64}$$

Therefore,

$$L\mathbf{x} = \lambda\mathbf{x}. \tag{1.65}$$

Thus, λ is an eigenvalue of L, with corresponding eigenvector $\mathbf{x}$. □

Definition 1.1.17 (Eigenvalues of a matrix)**.** Let M be an $n \times n$ matrix. The set of all eigenvalues of M is denoted by

$$\text{eigs}(M) = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}. \tag{1.66}$$

Here, the eigenvalues are listed with multiplicity.

Definition 1.1.18 (Definiteness of a matrix)**.** Let M be a real symmetric $n \times n$ matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then M is called

1. **Positive definite (PD)** if all eigenvalues are strictly positive, i.e.,

$$\lambda_i > 0, \quad \forall i = 1, 2, \ldots, n. \tag{1.67}$$

2. **Negative definite (ND)** if all eigenvalues are strictly negative, i.e.,

$$\lambda_i < 0, \quad \forall i = 1, 2, \ldots, n. \tag{1.68}$$

3. **Positive semidefinite (PSD)** if all eigenvalues are non-negative, i.e.,

$$\lambda_i \geq 0, \quad \forall i = 1, 2, \ldots, n. \tag{1.69}$$

4. **Negative semidefinite (NSD)** if all eigenvalues are non-positive, i.e.,

$$\lambda_i \leq 0, \quad \forall i = 1, 2, \ldots, n. \tag{1.70}$$

5. **Indefinite (ID)** if M has both positive and negative eigenvalues.

Example 1.1.19

Let

$$L \coloneqq \begin{bmatrix} -2 & 2 \\ 3 & -1 \end{bmatrix}. \tag{1.71}$$

We will find the eigenvalues and eigenvectors for L.

The eigenvalues are obtained by solving $\det(L - \lambda I) = 0$. Here,

$$\begin{aligned} \det(L - \lambda I) &= \begin{vmatrix} -2-\lambda & 2 \\ 3 & -1-\lambda \end{vmatrix} \\ &= (-2-\lambda)(-1-\lambda) - 6 \\ &= \lambda^2 + 3\lambda - 4 \\ &= (\lambda - 1)(\lambda + 4). \end{aligned} \tag{1.72}$$

Hence, the eigenvalues for L are $\lambda = 1$ and $\lambda = -4$.

For each eigenvalue, we find a non-zero vector $\mathbf{x} = [x_1, x_2]^T$ such that $(L - \lambda I)\mathbf{x} = 0$.

For $\lambda = 1$, we have

$$\begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{1.73}$$

Both equations give $3x_1 = 2x_2$, so $x_2 = \frac{3}{2}x_1$. Therefore, the eigenvectors corresponding to $\lambda = 1$ are precisely the vectors

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \frac{3}{2}x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ \frac{3}{2} \end{bmatrix}, \qquad x_1 \in \mathbb{R} \setminus \{0\}. \tag{1.74}$$

Choosing $x_1 = 2$, we obtain the eigenvector $\mathbf{x} = [2, 3]^T$.

For $\lambda = -4$, we have

$$\begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{1.75}$$

Both equations give $x_1 + x_2 = 0$, so $x_2 = -x_1$. Therefore, the eigenvectors corresponding to $\lambda = -4$ are precisely the vectors

$$\mathbf{x} = \begin{bmatrix} x_1 \\ -x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad x_1 \in \mathbb{R} \setminus \{0\}. \tag{1.76}$$

Choosing $x_1 = 1$, we obtain the eigenvector $\mathbf{x} = [1, -1]^T$.
Hence, $([2, 3]^T, 1)$ and $([1, -1]^T, -4)$ are eigenpairs for L.

Remark 1.1.20. Any nonzero scalar multiple of an eigenvector is also an eigenvector corresponding to the same eigenvalue.

Indeed, let $x \neq 0$ be an eigenvector of L corresponding to the eigenvalue λ. Then

$$Lx = \lambda x. \tag{1.77}$$

For any scalar $c \neq 0$, by linearity of L,

$$L(cx) = cLx = c\lambda x = \lambda(cx). \tag{1.78}$$

Hence, cx is also an eigenvector of L with the same eigenvalue λ.

2 Lecture 02

§2.1 Review on linear algebra—continued

Exercise 2.1.1. Find eigenvalues of the matrix

$$\begin{bmatrix} 6 & 0 \\ 0 & 7 \end{bmatrix}, \tag{2.1}$$

and classify it in terms of eigenvalues.

Answer 2.1.2. Let M denote the matrix in (2.1). The eigenvalues are obtained by solving $\det(M - \lambda I) = 0$. Here,

$$\begin{aligned} \det(M - \lambda I) &= \begin{vmatrix} 6-\lambda & 0 \\ 0 & 7-\lambda \end{vmatrix} \\ &= (6-\lambda)(7-\lambda). \end{aligned} \tag{2.2}$$

Therefore,

$$(6-\lambda)(7-\lambda) = 0 \iff \lambda = 6 \text{ or } \lambda = 7. \tag{2.3}$$

Hence, $\text{eigs}(M) = \{6, 7\}$. Since M is real symmetric and both eigenvalues are strictly positive, it is *positive definite (PD)* by Definition 1.1.18.

Remark 2.1.3. The eigenvalues of a diagonal matrix are precisely its diagonal entries.

Now consider the following matrix:

$$\begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}. \tag{2.4}$$

To find its eigenvalues, we solve

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 1-\lambda & -2 \\ 1 & -1-\lambda \end{vmatrix} \\ &= (1-\lambda)(-1-\lambda) + 2 \\ &= -1 - \lambda + \lambda + \lambda^2 + 2 \\ &= \lambda^2 + 1 = 0. \end{aligned}$$

Therefore,

$$\lambda^2 = -1 \implies \lambda = \pm i.$$

Its eigenvalues $\pm i$ are not real. Therefore, this matrix does not fall into any class given in Definition 1.1.18. But there are matrices that guaranteed to have real eigenvalues, which we discuss next. Recall:

Definition 2.1.4 (Transpose of a matrix)**.** Let A be an $m \times n$ matrix. Denote its entry whose row number is i and column number is j by $a_{i,j}$. Then transpose of the matrix A, denoted by A^T, is an $n \times m$ matrix whose entry at j-th row and i-th column is $a_{i,j}$. Symbolically, if

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \dots & a_{m,n} \end{bmatrix}, \tag{2.5}$$

then

$$A^T = \begin{bmatrix} a_{1,1} & a_{2,1} & \dots & a_{m,1} \\ a_{1,2} & a_{2,2} & \dots & a_{m,2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1,n} & a_{2,n} & \dots & a_{m,n} \end{bmatrix}. \tag{2.6}$$

Definition 2.1.5 (Symmetric matrix)**.** We call an $n \times n$ matrix A to be *symmetric* if

$$A = A^T. \tag{2.7}$$

Theorem 2.1.6
All the eigenvalues of a symmetric matrix A are real.

Proof. Will be done in the next class after developing some new idea. □

To prove this theorem, we now introduce new ideas in the context of vector spaces.

§2.1.1 Inner product and norm

It is generalisation of dot product of vectors that we learnt in the HSC/A-levels.

Example 2.1.7 (Motivating example)
Consider the $\mathbb{R}$-vector space $\mathbb{R}^2$. Define a function

$$\langle _, _ \rangle : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}, \tag{2.8}$$

given by

$$\langle _, _ \rangle (\mathbf{x}, \mathbf{y}) \coloneqq \mathbf{x}^T \mathbf{y}. \tag{2.9}$$

We will denote $\langle _, _ \rangle (\mathbf{x}, \mathbf{y})$ by $\langle \mathbf{x}, \mathbf{y} \rangle$.

Therefore, $\langle \mathbf{x}, \mathbf{y} \rangle$ is nothing but dot product of $\mathbf{x}$ and $\mathbf{y}$ in the high school sense, i.e.,

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x} \cdot \mathbf{y} \tag{2.10}$$

If $\mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^T$ and $\mathbf{y} = \begin{bmatrix} y_1 & y_2 \end{bmatrix}^T$, then

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x} \cdot \mathbf{y} = \sum_{k=1}^{2} x_k y_k. \tag{2.11}$$

We can then write length of a vector $\mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \in \mathbb{R}^2$, denoted by $\|\mathbf{x}\|$, in terms of dot product defined as above:

$$\|x\| = \sqrt{x_1^2 + x_2^2} = \sqrt{\langle x_1, x_2 \rangle}. \tag{2.12}$$

Furthermore, distance of $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$ can also be written in terms of the function $\langle _, _ \rangle$:

$$\begin{aligned}
&\sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} && (2.13)\\
=&\sqrt{\langle (x_1 - y_1, x_2 - y_2), (x_1 - y_1, x_2 - y_2) \rangle} && (2.14)\\
=&\sqrt{\langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle} && (2.15)\\
=&\|\mathbf{x} - \mathbf{y}\|. && (2.16)
\end{aligned}$$

Notice, we previously did not have the notion of length of a vector or distance between two vectors in $\mathbb{R}^n$, for arbitrary n. But the above definition of dot product can be generalised to any $\mathbb{R}^n$, and using that we can meaningfully introduce the notion of length and distance in arbitrary $\mathbb{R}^n$. This generalised notion of dot product and length will be called *inner product* and *norm*, respectively.

Definition 2.1.8 (Inner product)**.** An *inner product* on a vector space V over $\mathbb{R}$ is a function

$$\langle \cdot, \cdot \rangle : V \times V \longrightarrow \mathbb{R} \tag{2.17}$$

which assigns a real number $\langle u, v \rangle$ to every pair of vectors $u, v \in V$, satisfying the following properties for all $u, v, w \in V$ and $c \in \mathbb{R}$:

1. **Linearity in the first argument:**
$$\langle cu + v, w \rangle = c\langle u, w \rangle + \langle v, w \rangle. \tag{2.18}$$
2. **Symmetry:**
$$\langle u, v \rangle = \langle v, u \rangle. \tag{2.19}$$
3. **Positive definiteness:**
$$\langle u, u \rangle \geq 0, \tag{2.20}$$
with equality if and only if $u = 0$.

A vector space equipped with an inner product is called an *inner product space*.

Example 2.1.9

In the vector space $\mathbb{R}^n$ define

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y}. \tag{2.21}$$

Then it can be checked that the definition above defines an inner product on $\mathbb{R}^n$. Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ and $c \in \mathbb{R}$. We verify the properties of an inner product.

1. **Linearity in the first argument:**

$$\begin{aligned}
\langle c\mathbf{x} + \mathbf{y}, \mathbf{z} \rangle &= (c\mathbf{x} + \mathbf{y})^T \mathbf{z} \\
&= c\mathbf{x}^T \mathbf{z} + \mathbf{y}^T \mathbf{z}
\end{aligned}$$

$$= c\langle \mathbf{x}, \mathbf{z}\rangle + \langle \mathbf{y}, \mathbf{z}\rangle.$$

2. **Symmetry:**

$$\begin{aligned}\langle \mathbf{x}, \mathbf{y}\rangle &= \mathbf{x}^T\mathbf{y} \\ &= \sum_{i=1}^{n} x_i y_i \\ &= \sum_{i=1}^{n} y_i x_i \\ &= \mathbf{y}^T\mathbf{x} \\ &= \langle \mathbf{y}, \mathbf{x}\rangle.\end{aligned}$$

3. **Positive definiteness:**

$$\begin{aligned}\langle \mathbf{x}, \mathbf{x}\rangle &= \mathbf{x}^T\mathbf{x} \\ &= \sum_{i=1}^{n} x_i^2 \geq 0.\end{aligned}$$

Since $x_i^2 \geq 0$ for every i, equality holds if and only if $x_i = 0$ for all i, i.e., $\mathbf{x} = \mathbf{0}$.

Hence, the Euclidean dot product defines an inner product on $\mathbb{R}^n$.

Definition 2.1.10 (Norm induced by an inner product)**.** Suppose V is an $\mathbb{R}$-vector space with an inner product $\langle _, _\rangle$ defined on it. Then the function

$$\|.\| : V \to \mathbb{R} \tag{2.22}$$

defined by

$$\|\mathbf{x}\| \coloneqq \sqrt{\langle \mathbf{x}, \mathbf{x}\rangle} \tag{2.23}$$

is called *norm induced by this inner product.*

Example 2.1.11 (Euclidean norm on $\mathbb{R}^n$)

For $\mathbf{x} = (x_1, x_2, \ldots, x_n)^T \in \mathbb{R}^n$, the norm induced by the Euclidean inner product is

$$\begin{aligned}\|\mathbf{x}\|_2 &= \sqrt{\langle \mathbf{x}, \mathbf{x}\rangle} \\ &= \sqrt{\mathbf{x}^T\mathbf{x}} \\ &= \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.\end{aligned}$$

This is called the *Euclidean norm* or *2-norm* on $\mathbb{R}^n$.

Proposition 2.1.12

Norm induced by an inner product has some intrinsic properties:
For all $\mathbf{x}, \mathbf{y} \in V$ and $c \in \mathbb{R}$

1. **Positive definiteness:**
$$\|\mathbf{x}\| \geq 0, \tag{2.24}$$
and
$$\|\mathbf{x}\| = 0 \iff \mathbf{x} = 0. \tag{2.25}$$
2. **Homogeneity:**
$$\|c\mathbf{x}\| = |c|\,\|\mathbf{x}\|. \tag{2.26}$$
3. **Triangle inequality:**
$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|. \tag{2.27}$$

By making these properties definition, we can define general notion of norm that is independent from inner product.

Definition 2.1.13 (Norm on a vector space)**.** Let V be a real vector space. A *norm* on V is a function

$$\|\cdot\| : V \longrightarrow \mathbb{R}$$

that assigns a real number $\|v\|$ to each vector $v \in V$, satisfying the following properties for all $u, v \in V$ and all $c \in \mathbb{R}$:

1. **Non-negativity:**
$$\|v\| \geq 0. \tag{2.28}$$
2. **Definiteness:**
$$\|v\| = 0 \iff v = 0. \tag{2.29}$$
3. **Absolute homogeneity:**
$$\|cv\| = |c|\,\|v\|. \tag{2.30}$$
4. **Triangle inequality:**
$$\|u + v\| \leq \|u\| + \|v\|. \tag{2.31}$$

A real vector space V equipped with a norm is called a *normed vector space*.

Following example is particularly important for this course.

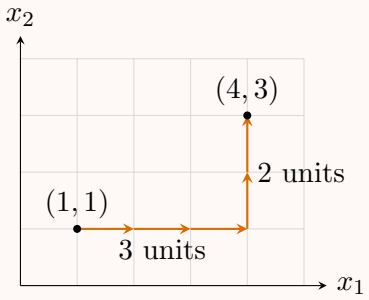
Example 2.1.14 (p-norm)

For $p \in \mathbb{R}$, define
$$\|.\|_p : \mathbb{R}^n \to \mathbb{R} \tag{2.32}$$
by
$$\|\mathbf{x}\|_p = \left(|x_1|^p + |x_2|^p + \cdots + |x_n|^p\right)^{\frac{1}{p}}, \tag{2.33}$$
where $\mathbf{x} = [x_1, \ldots, x_n]^T$.
We are not going to prove how this defines a norm on $\mathbb{R}^n$ though.
Putting $p = 2$ gives us the Euclidean norm of Example 2.1.11.

Putting $p = 1$ gives us

$$\|x\|_1 = |x_1| + |x_2| + \cdots + |x_n|, \tag{2.34}$$

which is also called "Manhattan norm" or "Taxicab" norm.
For example, consider the points[a] $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (4, 3)$ in $\mathbb{R}^2$.



The Manhattan distance between these points is

$$d_1(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_1 = |4 - 1| + |3 - 1| = 5.$$

Letting $p \to \infty$ gives

$$\|\mathbf{x}\|_p = \max(|x_1|, |x_2|, \ldots, |x_n|) \tag{2.35}$$

because if $x_i = \max(|x_1|, |x_2|, \ldots, |x_n|)$, then

$$\begin{aligned}
\lim_{p\to\infty} \|x\|_p &= \lim_{p\to\infty} (|x_i|^p)^{\frac{1}{p}} \left(\left|\frac{x_1}{x_i}\right|^p + \left|\frac{x_2}{x_i}\right|^p + \cdots + \left|\frac{x_i}{x_i}\right|^p + \cdots + \left|\frac{x_n}{x_i}\right|^p \right)^{\frac{1}{p}} \\
&= |x_i|(0 + 0 + \cdots + 1 + \cdots + 0)^0 \\
&= |x_i|.
\end{aligned}$$

[a] vectors in $\mathbb{R}^n$ are also referred as "points in $\mathbb{R}^n$"

Question 2.1.15. Let $\mathbf{x} = \begin{bmatrix} -6 \\ 4 \\ 0 \\ 1 \end{bmatrix}$. Find the 1-norm and 3-norm of $\mathbf{x}$.

Answer 2.1.16. Using the definition of the p-norm, we have
For $p = 1$,

$$\begin{aligned}
\|\mathbf{x}\|_1 &= |-6| + |4| + |0| + |1| \\
&= 6 + 4 + 0 + 1 \\
&= 11.
\end{aligned}$$

For $p = 3$,

$$\begin{aligned}
\|\mathbf{x}\|_3 &= \left(|-6|^3 + |4|^3 + |0|^3 + |1|^3\right)^{\frac{1}{3}} \\
&= (216 + 64 + 0 + 1)^{\frac{1}{3}} \\
&= \sqrt[3]{281}.
\end{aligned}$$

§2.1.2 Metric

Definition 2.1.17 (Metric)**.** Let V be a nonempty set. A function

$$d : V \times V \to \mathbb{R} \tag{2.36}$$

is called a *metric* if for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, the following properties hold:

1. **Positive definiteness:**
$$d(\mathbf{u}, \mathbf{v}) \geq 0, \tag{2.37}$$
with equality if and only if $\mathbf{u} = \mathbf{v}$.

2. **Symmetry:**
$$d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u}). \tag{2.38}$$

3. **Triangle inequality:**
$$d(\mathbf{u}, \mathbf{w}) \leq d(\mathbf{u}, \mathbf{v}) + d(\mathbf{v}, \mathbf{w}). \tag{2.39}$$

A set equipped with a metric is called a *metric space*.

Let V be a normed vector space. Then a metric can be induced by the norm as

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|. \tag{2.40}$$

In particular, if the norm is induced by an inner product, then

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{\langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle}. \tag{2.41}$$

Example 2.1.18 (Euclidean distance on $\mathbb{R}^n$)
For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the Euclidean distance is

$$d_2(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}. \tag{2.42}$$

For example, if $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (4, 3)$, then

$$\begin{aligned} d_2(\mathbf{x}, \mathbf{y}) &= \sqrt{(4-1)^2 + (3-1)^2} \\ &= \sqrt{13}. \end{aligned}$$

Definition 2.1.19 (Neighbourhood and ball)**.** Let d be a metric on $\mathbb{R}^n$. For $\mathbf{r}_0 \in \mathbb{R}^n$ and $a > 0$, the *open ball* of radius a centred at $\mathbf{r}_0$ is defined by

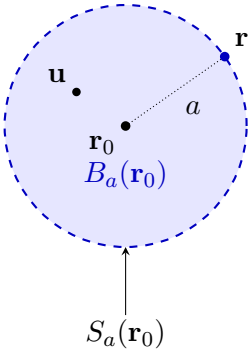
$$B_a(\mathbf{r}_0) = \{\mathbf{r} \in \mathbb{R}^n \mid d(\mathbf{r}, \mathbf{r}_0) < a\}. \tag{2.43}$$

It is also called the *a-neighbourhood* of $\mathbf{r}_0$.

The *sphere* of radius a centred at $\mathbf{r}_0$ is defined by

$$S_a(\mathbf{r}_0) = \{\mathbf{r} \in \mathbb{R}^n \mid d(\mathbf{r}, \mathbf{r}_0) = a\}. \tag{2.44}$$

In $\mathbb{R}^2$ with the Euclidean metric, the open ball consists of all points inside a circle, while the sphere is the circle itself. If $\mathbf{u} \in B_a(\mathbf{r}_0)$, then $d(\mathbf{u}, \mathbf{r}_0) < a$. If $\mathbf{r} \in S_a(\mathbf{r}_0)$, then $d(\mathbf{r}, \mathbf{r}_0) = a$.


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3 Lecture 03

§3.1 Review on linear algebra—continued

§3.1.1 Orthogonality

Definition 3.1.1 (Orthogonality)**.** Let V be an inner product space. Two vectors $\mathbf{x}, \mathbf{y} \in V$ are *orthogonal* if

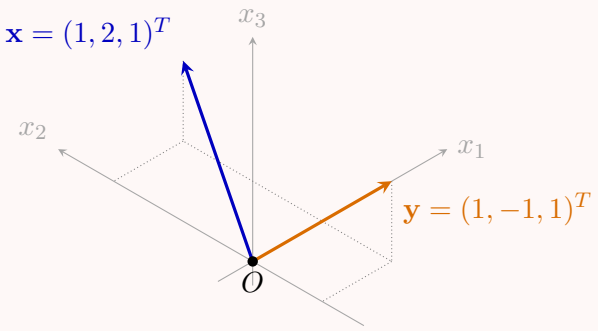
$$\langle \mathbf{x}, \mathbf{y} \rangle = 0. \tag{3.1}$$

Example 3.1.2

Let $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \in \mathbb{R}^3$. Then

$$\langle \mathbf{x}, \mathbf{y} \rangle = 1 \times 1 + 2 \times (-1) + 1 \times 1 = 0. \tag{3.2}$$

Therefore, $\mathbf{x}, \mathbf{y}$ are orthogonal/perpendicular.
The two vectors are shown below.



The vectors are perpendicular in $\mathbb{R}^3$, although their angle may appear different in this two-dimensional projection.

But since inner product is defined for any $\mathbb{R}^n$, we can talk about orthogonality there as well.

Example 3.1.3

Let $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \\ -7 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ 5 \\ 0 \end{bmatrix} \in \mathbb{R}^3$. Then

$$\langle \mathbf{x}, \mathbf{y} \rangle = 1 \times 1 + 2 \times (-1) + 1 \times 1 + 0 \times 5 + (-7) \times 0 = 0. \tag{3.3}$$

Therefore, $\mathbf{x}, \mathbf{y}$ are orthogonal.

Notice, $\mathbf{x}, \mathbf{y}$ are vectors in 5-dimensional space $\mathbb{R}^5$, which we can not visualise, yet we can talk about whether they are orthogonal/perpendicular to each other using

the language of (linear) algebra.

If the basis vectors $v_1, v_2, \ldots, v_n$ of the vector space $\mathbb{R}^n$ are pairwise orthogonal, i.e.,

$$\langle v_i, v_j \rangle = 0, \qquad i \neq j, \tag{3.4}$$

then the coefficients of any vector $v \in \mathbb{R}^n$ with respect to this basis can be determined easily. Write

$$v = \sum_{i=1}^{n} c_i v_i, \tag{3.5}$$

and taking the inner product with v_j on both sides, we obtain

$$\langle v, v_j \rangle = \left\langle \sum_{i=1}^{n} c_i v_i, v_j \right\rangle \tag{3.6}$$

$$= \sum_{i=1}^{n} c_i \langle v_i, v_j \rangle \tag{3.7}$$

$$= c_j \langle v_j, v_j \rangle, \tag{3.8}$$

since $\langle v_i, v_j \rangle = 0$ whenever $i \neq j$. Therefore,

$$c_j = \frac{\langle v, v_j \rangle}{\langle v_j, v_j \rangle} = \frac{\langle v, v_j \rangle}{\|v_j\|^2} \tag{3.9}$$

Thus, orthogonality allows us to determine each coefficient independently, without solving a system of linear equations. In particular, if the basis is orthonormal, then

$$v_j = \langle v, v_j \rangle \tag{3.10}$$

Example 3.1.4

Consider the following orthogonal basis of $\mathbb{R}^3$:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \qquad v_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}.$$

These vectors are pairwise orthogonal, with

$$\|v_1\|^2 = 2, \qquad \|v_2\|^2 = 2, \qquad \|v_3\|^2 = 4.$$

We want to express $v = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$ as $v = c_1 v_1 + c_2 v_2 + c_3 v_3$. Using (3.9),

$$c_1 = \frac{\langle v, v_1 \rangle}{\|v_1\|^2} = \frac{3+1}{2} = 2,$$

$$c_2 = \frac{\langle v, v_2 \rangle}{\|v_2\|^2} = \frac{3-1}{2} = 1,$$

$$c_3 = \frac{\langle v, v_3 \rangle}{\|v_3\|^2} = \frac{4 \cdot 2}{4} = 2.$$

Therefore,

$$v = 2v_1 + v_2 + 2v_3 = \begin{bmatrix}2\\2\\0\end{bmatrix} + \begin{bmatrix}1\\-1\\0\end{bmatrix} + \begin{bmatrix}0\\0\\4\end{bmatrix} = \begin{bmatrix}3\\1\\4\end{bmatrix}.$$

Each coefficient is obtained independently by taking an inner product with the corresponding basis vector.

Let V be a normed space.

Definition 3.1.5 (Normalised vectors)**.** A vector $v \in V$ is called *normalised* if

$$\|v\| = 1. \tag{3.11}$$

Suppose $v \in V$ is not normalised. But

$$\frac{v}{\|v\|} \tag{3.12}$$

is normalised. Because

$$\left\| \frac{v}{\|v\|} \right\| = {}^{1} \frac{1}{\|v\|} \|v\| = 1. \tag{3.13}$$

The process making a vector normalised in this way, is called *normalisation*.

Definition 3.1.6 (Orthogonal basis)**.** A set of basis $\{v_1, v_2, \ldots, v_n\}$ of a vector space V is called *orthogonal basis* if

$$\langle v_i, v_j \rangle = \begin{cases} c \in \mathbb{R} & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases} \tag{3.14}$$

Definition 3.1.7 (Orthonormal basis)**.** A set of basis $\{v_1, v_2, \ldots, v_n\}$ of a vector space V is called *orthonormal basis* if

$$\langle v_i, v_j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases} \tag{3.15}$$

Notation (Kronecker/Dirac delta). The Kronecker delta is defined by

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \tag{3.16}$$

For example,

$$\delta_{00} = 1, \qquad \delta_{11} = 1, \qquad \delta_{10} = 0.$$

Using this notation, the defining condition for an orthonormal basis $\{v_1, \ldots, v_n\}$ becomes

$$\langle v_i, v_j \rangle = \delta_{ij}.$$

In a sum, terms with unequal indices in the Kronecker delta vanish. The remaining terms have equal indices, so one summation can be removed.

[1] Using absolute homogeneity property of norm

Example 3.1.8

Let

$$\mathbf{x} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Consider

$$\sum_{i=1}^{2}\sum_{j=1}^{2} x_i \delta_{ij} x_j.$$

Since $\delta_{ij} = 0$ unless $i = j$,

$$\sum_{i=1}^{2}\sum_{j=1}^{2} x_i \delta_{ij} x_j = \sum_{i=1}^{2} x_i \cdot 1 \cdot x_i.$$

Expanding the original sum explicitly,

$$\begin{aligned}
\sum_{i=1}^{2}\sum_{j=1}^{2} x_i \delta_{ij} x_j &= x_1 \underbrace{\delta_{11}}_{1} x_1 + x_1 \underbrace{\delta_{12}}_{0} x_2 \\
&\quad + x_2 \underbrace{\delta_{21}}_{0} x_1 + x_2 \underbrace{\delta_{22}}_{1} x_2 \\
&= x_1^2 + x_2^2 \\
&= 1^2 + (-2)^2 = 5.
\end{aligned}$$

Example 3.1.9

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Here, A_{ij} denotes the entry in row i and column j. For example, $A_{11} = 1$ and $A_{21} = 3$.

Consider

$$\sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{2} A_{ij} \delta_{jk} x_k.$$

Only terms with $j = k$ remain. Replacing j by k, we obtain

$$\begin{aligned}
\sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{2} A_{ij} \delta_{jk} x_k &= \sum_{i=1}^{2}\sum_{k=1}^{2} A_{ik} x_k \\
&= A_{11}x_1 + A_{12}x_2 \\
&\quad + A_{21}x_1 + A_{22}x_2 \\
&= 1 \cdot 1 + 2 \cdot (-2) \\
&\quad + 3 \cdot 1 + 4 \cdot (-2) \\
&= 1 - 4 + 3 - 8 \\
&= -8.
\end{aligned}$$

Example 3.1.10

For the same matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},$$

consider

$$\sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{2} A_{ij}\delta_{ik}A_{jk}.$$

Only terms with $i = k$ remain. Replacing k by i, we obtain

$$\begin{aligned}\sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{2} A_{ij}\delta_{ik}A_{jk} &= \sum_{i=1}^{2}\sum_{j=1}^{2} A_{ij}A_{ji} \\ &= A_{11}A_{11} + A_{12}A_{21} \\ &\quad + A_{21}A_{12} + A_{22}A_{22} \\ &= 1^2 + 2\cdot 3 + 3\cdot 2 + 4^2 \\ &= 1 + 6 + 6 + 16 \\ &= 29.\end{aligned}$$

§3.1.2 Symmetric & skew symmetric matrix

Now we prove Theorem 2.1.6.

The proof below needs to be fixed.

Proof of Theorem 2.1.6. Let $\mathbf{x}$ be an eigenvector of A. Then $\mathbf{x} \neq 0$, and so $\|\mathbf{x}\| \neq 0$. Then

$$\langle A\mathbf{x}, A\mathbf{x}\rangle = \langle \lambda\mathbf{x}, \lambda\mathbf{x}\rangle \tag{3.17}$$

$$= \lambda^2\langle \mathbf{x}, \mathbf{x}\rangle \tag{3.18}$$

$$= \lambda^2\|x\|^2 \tag{3.19}$$

$$\implies \lambda = \sqrt{\frac{\langle A\mathbf{x}, A\mathbf{x}\rangle}{\|x\|^2}}. \tag{3.20}$$

Note that λ cannot be negative because numerator inside the square root (i.e., $\langle A\mathbf{x}, A\mathbf{x}\rangle$) is positive because norm is always positive. □

In addition,

Theorem 3.1.11

Eigenvectors with different eigenvalues of a symmetric matrix A are orthogonal.

Proof. Let A be a real symmetric matrix, and let $\mathbf{x}, \mathbf{y}$ be eigenvectors with distinct eigenvalues λ_1, λ_2. Thus,

$$A\mathbf{x} = \lambda_1\mathbf{x}, \qquad A\mathbf{y} = \lambda_2\mathbf{y}, \qquad \lambda_1 \neq \lambda_2.$$

On the one hand,

$$\langle A\mathbf{x}, \mathbf{y}\rangle = (\lambda_1\mathbf{x})^T\mathbf{y}$$

$$= \lambda_1 \mathbf{x}^T \mathbf{y}$$
$$= \lambda_1 \langle \mathbf{x}, \mathbf{y} \rangle.$$

On the other hand, since $A^T = A$,

$$\begin{aligned} \langle A\mathbf{x}, \mathbf{y} \rangle &= (A\mathbf{x})^T \mathbf{y} \\ &= \mathbf{x}^T A^T \mathbf{y} \\ &= \mathbf{x}^T A \mathbf{y} \\ &= \mathbf{x}^T (\lambda_2 \mathbf{y}) \\ &= \lambda_2 \langle \mathbf{x}, \mathbf{y} \rangle. \end{aligned}$$

Therefore,

$$(\lambda_1 - \lambda_2)\langle \mathbf{x}, \mathbf{y} \rangle = 0.$$

Since $\lambda_1 \neq \lambda_2$, we conclude that

$$\langle \mathbf{x}, \mathbf{y} \rangle = 0.$$

Hence, $\mathbf{x}$ and $\mathbf{y}$ are orthogonal. □

Example 3.1.12 (Construction of a symmetric matrix)

To construct a real symmetric matrix, choose its diagonal entries and the entries above the diagonal freely. Then reflect the entries above the diagonal across it, so that $a_{ij} = a_{ji}$.

For example, choosing the diagonal entries $5, 4, -3$ and the upper entries $7, -2, 13$ gives

$$S = \begin{bmatrix} 5 & 7 & -2 \\ 7 & 4 & 13 \\ -2 & 13 & -3 \end{bmatrix}.$$

Taking the transpose,

$$S^T = \begin{bmatrix} 5 & 7 & -2 \\ 7 & 4 & 13 \\ -2 & 13 & -3 \end{bmatrix} = S.$$

Hence, S is symmetric.

Definition 3.1.13 (Anti/skew symmetric matrix)**.** A matrix A is called *anti/skew symmetric* if

$$A = -A^T. \tag{3.21}$$

Example 3.1.14 (Construction of a skew-symmetric matrix)

For a real skew-symmetric matrix, $a_{ij} = -a_{ji}$. In particular, every diagonal entry satisfies

$$a_{ii} = -a_{ii} \implies a_{ii} = 0.$$

Thus, set the diagonal entries to zero, choose the entries above the diagonal freely, and place their negatives below the diagonal.

For example,

$$A = \begin{bmatrix} 0 & 7 & -23 \\ -7 & 0 & -13 \\ 23 & 13 & 0 \end{bmatrix}.$$

Taking the transpose,

$$A^T = \begin{bmatrix} 0 & -7 & 23 \\ 7 & 0 & 13 \\ -23 & -13 & 0 \end{bmatrix} = -A.$$

Hence, A is skew-symmetric.

Most matrices are neither symmetric nor anti symmetric. But given a matrix M, it can be written as sum of one symmetric and one anti symmetric matrix as follows:

$$M = S_m + A_M, \tag{3.22}$$

where

$$S_M = \frac{M + M^T}{2} \qquad \text{and} \qquad A_M = \frac{M - M^T}{2}. \tag{3.23}$$

§3.2 Review on multivariable calculus

In this course, we will optimise functions whose inputs and outputs may consist of several real numbers. A function with several input variables is called a *multivariable function*. The branch of mathematics concerned with such functions is known as *multivariable calculus*.

§3.2.1 Terminology Related to Functions

Definition 3.2.1. Let A and B be nonempty sets. A function $f : A \to B$ is a rule that assigns to every element $x \in A$ exactly one element $f(x) \in B$. Here,

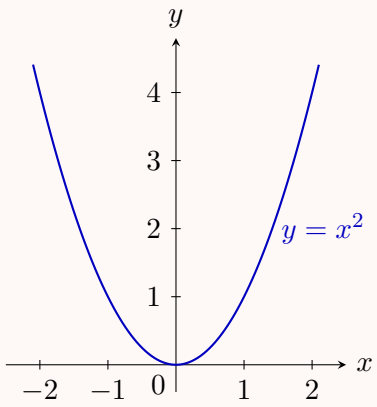
- A is called the *domain* of f, denoted by Dom f.
- B is called the *codomain* of f.
- The *range*, or *image*, of f is the subset of B consisting of all values actually attained by f, namely the set $f(A) = \{f(x) : x \in A\}$.
- The *graph* of f is the set $\text{Graph}(f) = \{(x, f(x)) : x \in A\}$.

Example 3.2.2

Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$.

Its domain and codomain are both $\mathbb{R}$, but range is $[0, \infty)$.

The graph of f consists of all points of the form (x, x^2), where $x \in \mathbb{R}$. A portion of this graph is shown below.

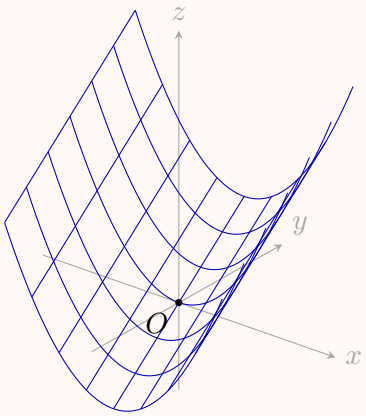


Graph of a function $f : \mathbb{R} \to \mathbb{R}$ can be represented in a plane. But for functions of several variables, the situation is more complicated. For example, the graph of a function $f : \mathbb{R}^2 \to \mathbb{R}$ lies in $\mathbb{R}^3$ and can often be visualized as a surface. However, the graph of a function $f : \mathbb{R}^2 \to \mathbb{R}^2$ lies in $\mathbb{R}^4$, which cannot be represented directly in ordinary three-dimensional space.

Example 3.2.3 (Graph of a function of two variables)
Consider $f : \mathbb{R}^2 \to \mathbb{R}$, defined by

$$f(x, y) = x^2 + y.$$

Its graph consists of the points $(x, y, x^2 + y)$ in $\mathbb{R}^3$. A portion of this surface is shown below.



Surface $z = x^2 + y$.

For each fixed y, the surface has a parabolic cross-section $z = x^2 + y$. For each fixed x, it has a straight-line cross-section with slope 1 in the y-direction.

§3.2.2 Directional Derivatives

We first recall the notion of differentiation from elementary calculus.

Definition 3.2.4. Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $a \in I$. We say that f is *differentiable at* a if the following limit exists as a finite real number:

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}. \tag{3.24}$$

The value $f'(a)$ is called the *derivative* of f at a.

Geometrically, the quotient $\frac{f(a+h)-f(a)}{h}$ represents the slope of the secant line joining $(a, f(a))$ and $(a+h, f(a+h))$. As h approaches zero, the secant slope approaches the slope of the tangent line, provided the limit exists.

Suppose now that $f : U \to \mathbb{R}$, where U is an open subset of $\mathbb{R}^n$ and $n \geq 2$.

Given $a \in U$, a small change in the input is represented by a vector $h \in \mathbb{R}^n$, rather than by a real number. One might attempt to imitate (3.24) by considering $\frac{f(a+h)-f(a)}{h}$.

However, this expression is not defined in the usual arithmetic of Euclidean spaces: its numerator is a real number, whereas its denominator is a vector. There is no canonical operation of division by a nonzero vector analogous to division by a nonzero real number.

There is also a geometric difficulty. In $\mathbb{R}$, a point can be approached only from the left or from the right. In $\mathbb{R}^n$, with $n \geq 2$, a point may be approached along infinitely many different directions. The rate of change of a function may depend on the particular direction in which its input is varied.

This leads us to the following definition:

Definition 3.2.5 (Directional derivative)**.** Let $U \subseteq \mathbb{R}^n$ be open, let $f : U \to \mathbb{R}^m$, and let $a \in U$. For a nonzero vector $v \in \mathbb{R}^n$, the *directional derivative* of f at a in the direction v is defined by

$$D_v f(a) = \lim_{t \to 0} \frac{f(a+tv) - f(a)}{t}, \tag{3.25}$$

provided that this limit exists in $\mathbb{R}^m$.

The directional derivative $D_v f(a)$ measures the instantaneous rate of change of f at a as the input moves along the line $a + tv$. When v is a unit vector, it measures the rate of change per unit distance in that direction.

Remark 3.2.6. The ordinary derivative is a special case of the directional derivative. Indeed, when $n = m = 1$ and $v = 1$, the definition reduces to $D_1 f(a) = f'(a)$.

4 Lecture 04

In the previous lecture, we started reviewing multivariable calculus. We will return to it when we study multivariable optimisation. For now, we introduce some terminology and study optimisation of smooth functions of one variable.

§4.1 Terminology related to optimisation

Our basic minimisation problem is

$$\min_{\mathbf{x}\in S} f(\mathbf{x}), \qquad f : \operatorname{Dom} f \to \mathbb{R}, \qquad \varnothing \neq S \subseteq \operatorname{Dom} f \subseteq \mathbb{R}^n. \tag{4.1}$$

Here, we choose an input from S so that the value of f is as small as possible. Notice that S need not be the entire domain of f.

Definition 4.1.1 (Objective function and feasible set)**.** The function f that we want to optimise is called the *objective function*. The set S of allowed inputs is called the *feasible set*. An input $\mathbf{x} \in S$ is called a *feasible point*.

Example 4.1.2

Suppose x_1 and x_2 represent quantities of nuts and bolts, and their cost is given by the function

$$f(x_1, x_2) = 5x_1^2 + x_2^2.$$

The formula makes sense for every $(x_1, x_2) \in \mathbb{R}^2$. However, negative quantities do not make sense in this context. Thus, we may use the feasible set

$$S = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq 0,\ x_2 \geq 0\}.$$

The feasible set records the restrictions on the inputs; the formula for f alone does not.

Definition 4.1.3 (Optimiser and optimum value)**.** A point $\mathbf{x}^* \in S$ satisfying

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \qquad \text{for every } \mathbf{x} \in S \tag{4.2}$$

is called a *minimiser*. The corresponding value $f(\mathbf{x}^*)$ is called the *minimum value*. For maximisation, the inequality is reversed, and we use the terms *maximiser* and *maximum value*. In general, the input $\mathbf{x}^*$ is an *optimiser*, whereas the output $f(\mathbf{x}^*)$ is an *optimum value*.

The distinction is important: an optimiser is a point in the feasible set, while an optimum value is a real number. Also, a feasible point is not necessarily an optimiser; it must satisfy the required comparison.

§4.1.1 Local and global extrema

A point may give a smaller value than all nearby feasible points without giving the smallest value over the whole feasible set.

Definition 4.1.4 (Local and global minimiser)**.** Let $f : \operatorname{Dom} f \to \mathbb{R}$ and $S \subseteq \operatorname{Dom} f$. A point $\mathbf{x}^* \in S$ is a *local minimiser* of f over S if there exists $\delta > 0$ such that

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \qquad \text{for all } \mathbf{x} \in S \text{ with } \|\mathbf{x} - \mathbf{x}^*\| < \delta. \tag{4.3}$$

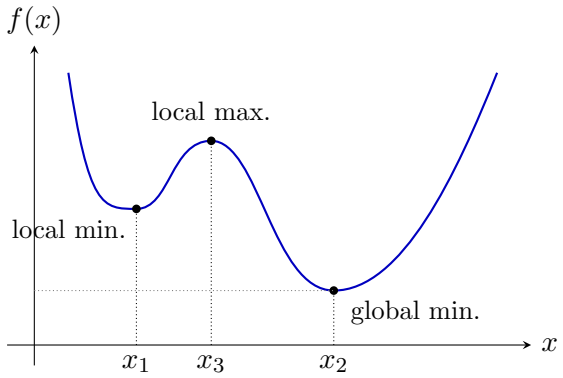
It is a *global minimiser* if the inequality holds for every $\mathbf{x} \in S$, as in (4.2). The corresponding function values are called a *local minimum* and the *global minimum*, respectively.

Definition 4.1.5 (Local and global maximiser)**.** A point $\mathbf{x}^* \in S$ is a *local maximiser* if there exists $\delta > 0$ such that

$$f(\mathbf{x}^*) \geq f(\mathbf{x}) \qquad \text{for all } \mathbf{x} \in S \text{ with } \|\mathbf{x} - \mathbf{x}^*\| < \delta. \tag{4.4}$$

It is a *global maximiser* if the inequality holds for every $\mathbf{x} \in S$. Minima and maxima are collectively called *extrema*; minimum and maximum values are *optimum values*.

The following schematic shows two local minimisers and one local maximiser.

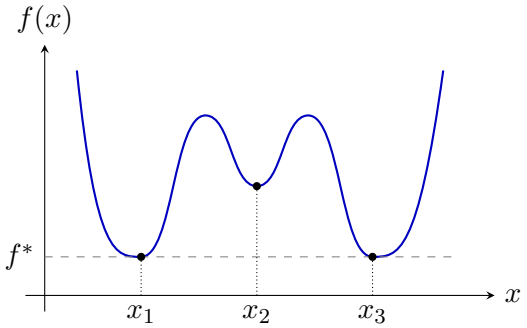


Every global minimiser is a local minimiser, but the converse need not hold. Knowing the behaviour near x_1 does not tell us whether a lower valley exists elsewhere. This is why local information alone usually does not establish global optimality. In general, finding global minima is a hard problem.

There may be several local minimisers and several global minimisers. However, if a global minimum is attained, its *value* is unique. Indeed, if $\mathbf{x}_1^*$ and $\mathbf{x}_2^*$ are global minimisers, then

$$f(\mathbf{x}_1^*) \leq f(\mathbf{x}_2^*) \quad \text{and} \quad f(\mathbf{x}_2^*) \leq f(\mathbf{x}_1^*).$$

Therefore, $f(\mathbf{x}_1^*) = f(\mathbf{x}_2^*)$. The inputs can be different even though their outputs agree.



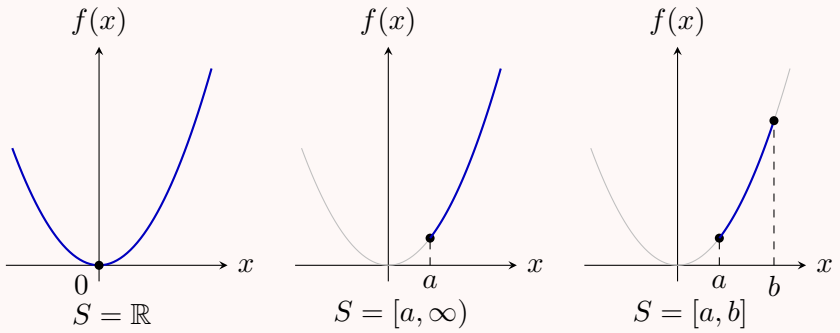
Three local minimisers; x_1 and x_3 share the global minimum f^*.

§4.1.2 The role of the feasible set

Changing the feasible set can change both the location and the existence of an optimum, even when the objective function stays exactly the same.

Example 4.1.6 (The same function on three feasible sets)
Consider $f(x) = x^2$ and choose $0 < a < b$.

1. If $S = \mathbb{R}$, the global minimiser is $x^* = 0$ and the minimum value is 0. There is no local or global maximum.
2. If $S = [a, \infty)$, then 0 is infeasible. Since x^2 is strictly increasing for $x > 0$, the global minimiser is a and the minimum value is a^2. There is still no global maximum.
3. If $S = [a, b]$, the global minimiser is a, with value a^2, and the global maximiser is b, with value b^2.



The blue portions correspond to feasible inputs; $0 < a < b$.

In the terminology used in class, a *natural extremum* comes from the behaviour of the function itself, without an added restriction on its domain. An *induced extremum* arises because a constraint removes inputs that would otherwise prevent that point from being an extremum.

For $f(x) = x^2$, the minimum at 0 over $\mathbb{R}$ is natural. The minimum at $a > 0$ over $[a, \infty)$ is induced by the constraint. Over $[a, b]$, with $0 < a < b$, both the minimum at a and the maximum at b are induced. They are also local extrema relative to the feasible set.

If the feasible interval includes 0, the natural minimiser remains feasible.

Definition 4.1.7 (Constrained and unconstrained optimisation)**.** If we restrict the allowed inputs beyond the domain of the function, we have a *constrained optimisation problem*. If $S = \operatorname{Dom} f$, so that no additional restriction is imposed, we have an *unconstrained optimisation problem*.

For most of the initial part of this course, we will study unconstrained problems. Constraints will be considered later. The derivative tests below concern interior points where movement in both directions is possible.

§4.2 Unconstrained optimisation of smooth single-variable functions

We now consider functions $f : \mathbb{R} \to \mathbb{R}$, or more generally $f : I \to \mathbb{R}$ on an open interval I, with no additional constraints.

Definition 4.2.1 (Smooth function)**.** A function on an open interval is called *smooth* if it has continuous derivatives of every order; that is, it can be differentiated infinitely many times.

Example 4.2.2

Polynomials, e^x, $\sin x$ and $\cos x$ are smooth on $\mathbb{R}$. A polynomial is still infinitely differentiable: after sufficiently many derivatives we obtain the zero function, which can be differentiated again. Functions such as $1/x$ and $\tan x$ are smooth on open intervals within their domains, but are not defined on all of $\mathbb{R}$.

Non-example 4.2.3

The function $|x|$ is not differentiable at 0.

§4.2.1 First and second derivative tests

Theorem 4.2.4 (First-order necessary condition)

Let x^* be an interior point of the domain of f, and suppose f is differentiable at x^*. If x^* is a local minimiser or a local maximiser, then

$$f'(x^*) = 0. \tag{4.5}$$

For the smooth functions considered here, we find the *critical points* by solving $f'(x) = 0$. Each is only a candidate for a local optimiser and must still be checked.

Theorem 4.2.5 (Second derivative test)

Suppose f has continuous first and second derivatives near an interior point x^*, and $f'(x^*) = 0$.

1. If $f''(x^*) > 0$, then x^* is a local minimiser.
2. If $f''(x^*) < 0$, then x^* is a local maximiser.
3. If $f''(x^*) = 0$, this test is inconclusive.

Example 4.2.6 (The test works for x^2)

Let $f(x) = x^2$. Then

$$f'(x) = 2x = 0 \iff x^* = 0, \qquad f''(x) = 2, \qquad f''(0) = 2 > 0.$$

Thus, 0 is a local minimiser. In fact, $x^2 \geq 0 = f(0)$ for all $x \in \mathbb{R}$, so 0 is also the

unique global minimiser.

Example 4.2.7 (The test is inconclusive for x^4)
Let $f(x) = x^4$. Then

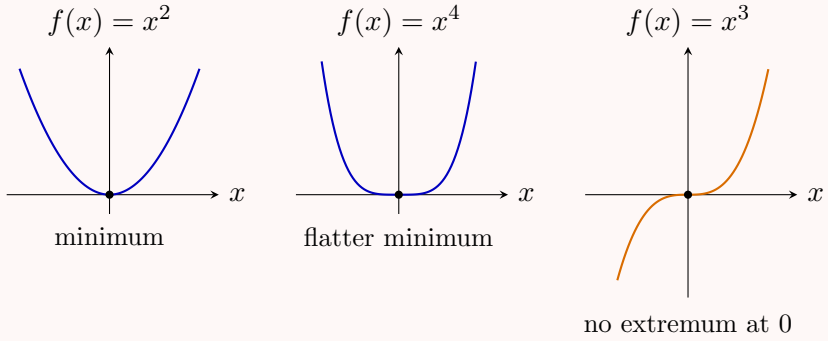
$$f'(x) = 4x^3 = 0 \iff x^* = 0, \qquad f''(x) = 12x^2, \qquad f''(0) = 0.$$

The second derivative test gives no conclusion. Nevertheless, $x^4 > 0 = f(0)$ for every $x \neq 0$, so 0 is the unique global minimiser. A zero second derivative does not mean there is no extremum.

Example 4.2.8 (Zero first derivative does not guarantee an extremum)
Let $f(x) = x^3$. Then

$$f'(x) = 3x^2 = 0 \iff x^* = 0, \qquad f''(x) = 6x, \qquad f''(0) = 0.$$

For every $h > 0$, we have $f(-h) < f(0) < f(h)$. Thus, 0 is neither a local minimiser nor a local maximiser. The same pair of values $f'(0) = f''(0) = 0$ occurs for x^4, but the behaviour is different.

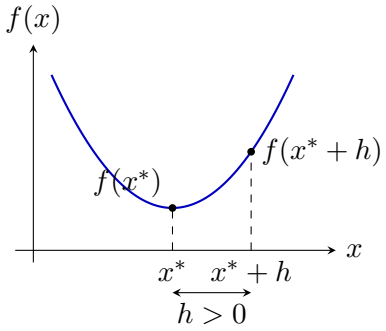


§4.2.2 Taylor expansion and higher derivative test

To understand these tests, compare the value at x^* with the value at a nearby point $x^* + h$. A positive h moves to the right; a negative h moves to the left. At a local minimum, we require

$$f(x^* + h) - f(x^*) \geq 0 \qquad \text{for all sufficiently small } |h|. \tag{4.6}$$

At a local maximum, the difference is nonpositive.



A function is *analytic* near a point if it agrees with its Taylor series there. Smoothness alone does not guarantee this. As in the lecture, assume here that f is analytic near x^*, so that we can write

$$f(x^* + h) = f(x^*) + hf'(x^*) + \frac{h^2}{2!}f''(x^*) + \frac{h^3}{3!}f^{(3)}(x^*) + \cdots . \tag{4.7}$$

Here, $f^{(k)}(x^*)$ denotes the kth derivative evaluated at x^*. Each of these evaluated derivatives is a fixed number; h is the quantity that varies.

Why must the first derivative vanish? Subtracting $f(x^*)$ from the Taylor expansion gives

$$f(x^* + h) - f(x^*) = hf'(x^*) + \frac{h^2}{2!}f''(x^*) + \cdots .$$

For sufficiently small $|h|$, the first nonzero term determines the sign of this difference. If $f'(x^*) \neq 0$, the leading term $hf'(x^*)$ changes sign when we move from the right of x^* to the left. The difference then cannot stay nonnegative on both sides, as a local minimum requires. It cannot stay nonpositive on both sides either. Therefore, at a local optimiser we must have $f'(x^*) = 0$.

Why does the second derivative classify the point? When $f'(x^*) = 0$, the expansion starts with

$$f(x^* + h) - f(x^*) = \frac{h^2}{2!}f''(x^*) + \frac{h^3}{3!}f^{(3)}(x^*) + \cdots . \tag{4.8}$$

If $f''(x^*) \neq 0$, the quadratic term determines the sign for sufficiently small nonzero h. Since h^2 is positive on both sides, $f''(x^*) > 0$ makes the nearby values larger, giving a local minimum. Similarly, $f''(x^*) < 0$ makes them smaller, giving a local maximum.

What if the second derivative also vanishes? If $f'(x^*) = f''(x^*) = 0$, we look at the next terms:

$$f(x^* + h) - f(x^*) = \frac{h^3}{3!}f^{(3)}(x^*) + \frac{h^4}{4!}f^{(4)}(x^*) + \cdots . \tag{4.9}$$

If $f^{(3)}(x^*) \neq 0$, the leading cubic term changes sign across x^*, so there is no local minimum or maximum. If $f^{(3)}(x^*) = 0$ but $f^{(4)}(x^*) \neq 0$, the fourth-order term comes first. Since $h^4 > 0$ for $h \neq 0$, a positive $f^{(4)}(x^*)$ gives a local minimum and a negative value gives a local maximum. If this derivative also vanishes, we look at the next order. The rule below applies once a first nonzero derivative is found.

Theorem 4.2.9 (First nonzero derivative test)

Let f be analytic near an interior point x^*, as assumed in the Taylor argument above. Suppose, for some integer $m \geq 1$,

$$f^{(k)}(x^*) = 0 \quad (1 \leq k < m), \qquad f^{(m)}(x^*) \neq 0. \tag{4.10}$$

Then:

1. If m is odd, x^* is neither a local minimiser nor a local maximiser.
2. If m is even and $f^{(m)}(x^*) > 0$, then x^* is a local minimiser.
3. If m is even and $f^{(m)}(x^*) < 0$, then x^* is a local maximiser.

Proof. In the Taylor argument above, the terms before order m vanish. Thus, the first nonzero term in $f(x^* + h) - f(x^*)$ is

$$\frac{h^m}{m!} f^{(m)}(x^*).$$

For sufficiently small nonzero h, this term determines the sign. If m is odd, h^m changes sign between the two sides, so there can be no local minimum or maximum. If m is even, h^m is positive on both sides. The sign is then the sign of $f^{(m)}(x^*)$: positive gives a local minimum, and negative gives a local maximum. □

Once the first nonzero derivative is found, later derivatives do not affect this classification; they need not vanish. In particular, the test does not require every odd-order derivative to be zero at a local optimum.

Returning to $f(x) = x^4$, we have $f^{(3)}(x) = 24x$ and $f^{(4)}(x) = 24$. At 0, the first nonzero evaluated derivative has even order 4 and is positive, confirming a local minimum. For $f(x) = x^3$, $f^{(3)}(0) = 6 \neq 0$ has odd order, so 0 is neither a local minimum nor a local maximum.

Remark 4.2.10 (The test need not finish for every smooth function)**.** Being smooth does not guarantee that a first nonzero derivative will be found. For example, the smooth function

$$q(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0 \end{cases}$$

has $q^{(k)}(0) = 0$ for every $k \geq 1$. Nevertheless, 0 is a global minimiser, since $q(0) = 0$ and $q(x) > 0$ for $x \neq 0$. Thus, if all evaluated derivatives vanish, the higher derivative test gives no conclusion; we need another argument.

Example 4.2.11 (Cubic function)

Find the local minima and maxima of

$$f(x) = x^3 - 6x^2 + 11x - 6, \qquad x \in \mathbb{R}. \tag{4.11}$$

Solution. First, note that

$$f(x) = (x - 1)(x - 2)(x - 3).$$

Thus, the graph crosses the x-axis at 1, 2 and 3. To locate the possible local optimisers, we solve $f'(x) = 0$.

Step 1: Find the candidates. Differentiating gives

$$f'(x) = 3x^2 - 12x + 11.$$

Hence,

$$\begin{aligned} 3x^2 - 12x + 11 &= 0, \\ x &= \frac{12 \pm \sqrt{144 - 132}}{6} \\ &= \frac{12 \pm \sqrt{12}}{6} = 2 \pm \frac{\sqrt{3}}{3} = 2 \pm \frac{1}{\sqrt{3}}. \end{aligned} \tag{4.12}$$

Following the board notation, let

$$x_1^* = 2 + \frac{1}{\sqrt{3}}, \qquad x_2^* = 2 - \frac{1}{\sqrt{3}}.$$

Notice that $x_2^* < x_1^*$.

Step 2: Classify each candidate separately. We have $f''(x) = 6x - 12$. Therefore,

$$f''(x_1^*) = 6\left(2 + \frac{1}{\sqrt{3}}\right) - 12 = \frac{6}{\sqrt{3}} = 2\sqrt{3} > 0,$$
$$f''(x_2^*) = 6\left(2 - \frac{1}{\sqrt{3}}\right) - 12 = -\frac{6}{\sqrt{3}} = -2\sqrt{3} < 0.$$

Thus, x_1^* is a local minimiser and x_2^* is a local maximiser. Both classifications stop at the second derivative because it is already nonzero at each point.

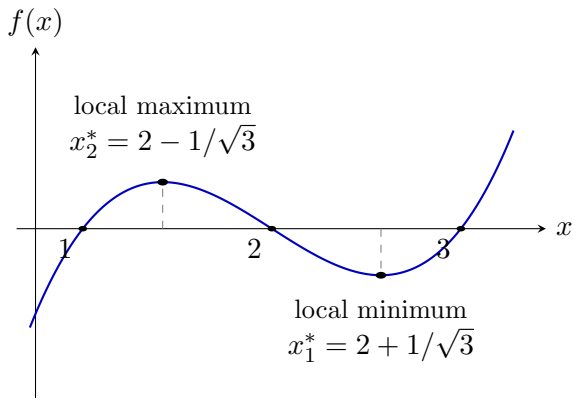
Step 3: Find the corresponding optimum values. Set $t = x - 2$. Then

$$\begin{aligned} f(2+t) &= (2+t)^3 - 6(2+t)^2 + 11(2+t) - 6 \\ &= (8 + 12t + 6t^2 + t^3) - (24 + 24t + 6t^2) + 22 + 11t - 6 \\ &= t^3 - t. \end{aligned}$$

Consequently,

$$f(x_1^*) = \left(\frac{1}{\sqrt{3}}\right)^3 - \frac{1}{\sqrt{3}} = -\frac{2}{3\sqrt{3}},$$
$$f(x_2^*) = \left(-\frac{1}{\sqrt{3}}\right)^3 + \frac{1}{\sqrt{3}} = \frac{2}{3\sqrt{3}}.$$

These are the local minimum and local maximum *values*, respectively.



Step 4: Check whether the extrema are global. Since the leading term is x^3,

$$\lim_{x\to\infty} f(x) = \infty, \qquad \lim_{x\to-\infty} f(x) = -\infty.$$

Thus, on $S = \mathbb{R}$ the function has neither a global minimiser nor a global maximiser. The two points found above give only local extrema. As a check, $f'(x) = 3(x - x_2^*)(x - x_1^*)$ is positive before x_2^*, negative between x_2^* and x_1^*, and positive after x_1^*. The graph therefore increases, decreases, and then increases again, consistent with the classifications above.

Different candidate points may require different numbers of derivatives. At each candidate, we stop at the first nonzero *evaluated* derivative; the derivative function need not be identically zero.

Finally, interior differentiability is essential for the necessary condition $f'(x^*) = 0$. For example, $|x|$ has a minimum at 0, where the derivative does not exist. Similarly, x^2 on $[a, \infty)$ with $a > 0$ has its minimum at a, although $f'(a) = 2a \neq 0$. At a constrained boundary, we cannot move in both directions while remaining feasible, so the argument for (4.5) does not apply. We will study the appropriate conditions for constrained optimisation later.

5 Lecture 05

§5.1 Unconstrained optimisation of smooth single-variable functions—continued

§5.1.1 Taylor expansion and higher derivative test—continued

Based on the theorems developed in the last lecture, we will workout on another example.

Exercise 5.1.1. Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = (5x^5 + 3x^3)e^x. \tag{5.1}$$

Find its local maximisers/minimizers, and their local maxima/minima.

Answer 5.1.2. Applying the product rule to Equation 5.1, we get

$$\begin{aligned} f'(x) &= (25x^4 + 9x^2)e^x + (5x^5 + 3x^3)e^x \\ &= x^2e^x(5x^3 + 25x^2 + 3x + 9). \end{aligned} \tag{5.2}$$

As $e^x > 0$ for every $x \in \mathbb{R}$, by Theorem 4.2.4, potential local minimizers (aka. critical points) are determined by the equation

$$x^2(5x^3 + 25x^2 + 3x + 9) = 0. \tag{5.3}$$

Thus, $x = 0$ is a critical point. The remaining critical points satisfy

$$5x^3 + 25x^2 + 3x + 9 = 0. \tag{5.4}$$

The cubic polynomial in Equation 5.4 has exactly one real root. Numerically, this root is

$$\alpha \approx -4.952238165.$$

Hence, the function has two distinct critical points, namely $x = \alpha$ and $x = 0$.

To classify these points, we compute the second derivative:

$$f''(x) = xe^x(5x^4 + 50x^3 + 103x^2 + 18x + 18). \tag{5.5}$$

At $x = \alpha$,

$$f''(\alpha) \approx 21.3643 > 0.$$

Hence, by Theorem 4.2.5, α is a local minimizer. The corresponding local minimum is

$$f(\alpha) \approx -107.831366.$$

At $x = 0$,

$$f'(0) = 0, \qquad f''(0) = 0.$$

Thus, the second-order test is inconclusive. We therefore apply Theorem 4.2.9:

Differentiating Equation 5.5, we get

$$f'''(x) = e^x(5x^5 + 75x^4 + 303x^3 + 327x^2 + 72x + 18). \tag{5.6}$$

In particular,

$$f'''(0) = 18 \neq 0.$$

Therefore,

$$f'(0) = f''(0) = 0, \qquad f'''(0) \neq 0.$$

So first nonzero derivative at $x = 0$ is of order $m = 3$. Since m is odd, $x = 0$ is neither a local minimiser nor a local maximiser.

We conclude that there is no local maxima of f, and it a has local minimizer at $x^* \approx -4.952238165$ and corresponding minimum is $f(x^*) \approx -107.831366$.

6 Lecture 06

§6.1 Review on multivariable calculus (continuation)

§6.1.1 Multivariable chain rule

We pickup where we left in section 3.2. This time our goal is to derive some useful formula to compute directional derivatives, so that we don't have to use Equation 3.25 each time to evaluate directional derivatives (just like summation, product, chain rule in single variable calculus).

Theorem 6.1.1 (Multivariable chain rule)

Let $I \subseteq \mathbb{R}$ be an open interval, $U \subseteq \mathbb{R}^n$ be open, and $u = (u_1, \dots, u_n) : I \to U$ and $f : U \to \mathbb{R}$ be functions. Suppose that u is differentiable at $t_0 \in I$ and f is differentiable at $u(t_0)$. Then $f \circ u$ is differentiable at t_0, and

$$(f \circ u)'(t_0) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(u(t_0))u_i'(t_0). \tag{6.1}$$

This theorem is best understood by practicing some examples.

Example 6.1.2

Let

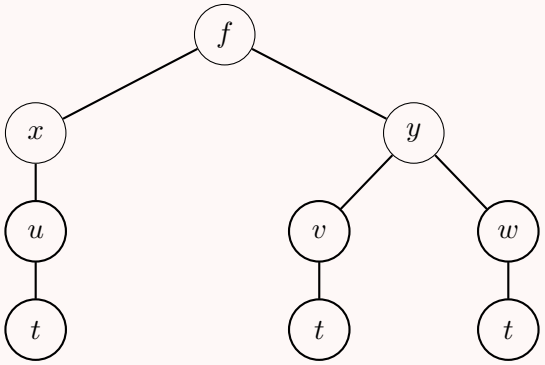
$$f(x, y) = x + y, \tag{6.2}$$

where

$$\begin{aligned} x &= e^u & y &= vw, \\ u &= t^2 & v &= e^t, \\ & & w &= \sin t. \end{aligned} \tag{6.3}$$

We wish to compute $\frac{df}{dt}$ using the multivariable chain rule.

The following mental picture is useful for efficiently applying the multivariable chain rule: Starting from f, follow every possible path to t. Along each path, multiply the corresponding partial derivatives, and then add the contributions from all paths.



Therefore, the chain rule gives

$$\frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{du}\frac{du}{dt} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial v}\frac{dv}{dt} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial w}\frac{dw}{dt}. \tag{6.4}$$

Now,

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 1, \frac{dx}{du} = e^u, \frac{\partial y}{\partial v} = w, \frac{\partial y}{\partial w} = v, \frac{du}{dt} = 2t, \frac{dv}{dt} = e^t, \frac{dw}{dt} = \cos t.$$

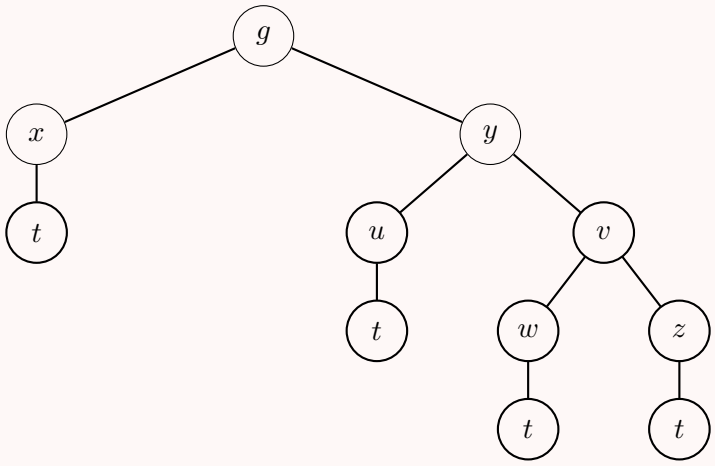
Substituting everything into Equation 6.4, we get

$$\frac{df}{dt} = e^u(2t) + w(e^t) + v(\cos t) = 2te^{t^2} + e^t \sin t + e^t \cos t. \tag{6.5}$$

To get used to using chain rule from the dependency tree, let's do another example.

Example 6.1.3

Suppose we have a function whose variable dependency tree looks as follows:



Then

$$\frac{dg}{dt} = \frac{\partial g}{\partial x}\frac{dx}{dt} + \frac{\partial g}{\partial y}\frac{\partial y}{\partial u}\frac{du}{dt} + \frac{\partial g}{\partial y}\frac{\partial y}{\partial v}\frac{\partial v}{\partial w}\frac{dw}{dt} + \frac{\partial g}{\partial y}\frac{\partial y}{\partial v}\frac{\partial v}{\partial z}\frac{dz}{dt}. \tag{6.6}$$

Now we are ready to prove a result that will help us to easily calculate directional derivatives.

Proposition 6.1.4 (Characterisation of directional derivatives)

Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}$ be differentiable at $p \in U$. Then, for every $v = (v_1, \ldots, v_n) \in \mathbb{R}^n$, the directional derivative of f at $p = (p_1, \ldots, p_n)$ in the direction v exists and is given by

$$D_v f(p) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(p) v_i. \tag{6.7}$$

Proof. Since U is open, we can choose $\varepsilon > 0$ sufficiently small so that $p + tv \in U$ whenever $|t| < \varepsilon$.

Consider the curve

$$u : (-\varepsilon, \varepsilon) \to U, \qquad u(t) = p + tv = \begin{bmatrix} p_1 + tv_1 \\ p_2 + tv_2 \\ \vdots \\ p_n + tv_n \end{bmatrix}.$$

Let $u_i(t) = p_i + tv_i$. Here $u(0) = p$ and $u_i'(t) = v_i$ for $i = 1, \ldots, n$. By the definition of the directional derivative,

$$D_v f(p) = \lim_{t \to 0} \frac{f(p + tv) - f(p)}{t} = \frac{d}{dt} f(u(t)) \bigg|_{t=0}.$$

Applying the Multivariable chain rule, we get

$$\begin{aligned} D_v f(p) &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(u(0)) u_i'(0) \\ &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(p) v_i, \end{aligned}$$

which complete the proof. □

The importance of this proposition is that we can now calculate directional derivatives using ordinary partial derivatives, without returning to the limit definition each time.

Example 6.1.5

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by

$$f(x_1, x_2) = x_1^2 + 3x_2.$$

We wish to compute the directional derivative of f at $x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ in the direction $v = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$.

By Proposition 6.1.4,

$$D_v f(x_0) = \sum_{i=1}^{2} \frac{\partial f}{\partial x_i}(x_0) v_i.$$

Now,

$$\frac{\partial f}{\partial x_1} = 2x_1, \qquad \frac{\partial f}{\partial x_2} = 3.$$

Therefore,

$$\begin{aligned} D_v f(x_0) &= \frac{\partial f}{\partial x_1}(x_0) v_1 + \frac{\partial f}{\partial x_2}(x_0) v_2 \\ &= 2(1)(1) + 3(-3) \\ &= -7. \end{aligned}$$

§6.1.2 The gradient

Definition 6.1.6 (Gradient)**.** Let $f : U \to \mathbb{R}$ be differentiable at $p \in U$, where $U \subseteq \mathbb{R}^n$ is open. The *gradient* of f at p is the vector

$$\nabla f(p) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(p) \\ \vdots \\ \frac{\partial f}{\partial x_n}(p) \end{bmatrix}. \tag{6.8}$$

With this notation, the result of the previous proposition becomes

$$D_v f(x_0) = \langle \nabla f(x_0), v \rangle. \tag{6.9}$$

Thus, once the gradient at a point is known, we can calculate the directional derivative along any direction by taking an inner product.

§6.1.3 Geometric interpretation of extrema

From single-variable calculus, if a differentiable function has a local minimum at an interior point x_0, then its derivative at that point must vanish, i.e., $f'(x_0) = 0$.

Geometrically, at a local minimum, moving slightly to either the left or the right cannot decrease the value of the function. Similarly, at a local maximum, moving in either direction cannot increase its value.

Insert a figure showing moving slightly left of right from the maxima/minima increment the function value (optional)

We wish to generalise this observation to functions of several variables. Suppose $f : \mathbb{R}^n \to \mathbb{R}$ has a local minimum at x_0. Then, for every sufficiently small displacement $h \in \mathbb{R}^n$, we must have

$$f(x_0 + h) - f(x_0) \geq 0. \tag{6.10}$$

In other words, moving in any direction cannot decrease the function value. For a local maximum, the inequality is reversed.

Add a $\mathbb{R}^2 \to \mathbb{R}$ function to show a local maxima.

Proposition 6.1.7 (Necessary condition for a local extremum)

Let $U \subseteq \mathbb{R}^n$ be open and $f : U \to \mathbb{R}$ be differentiable at $x_0 \in U$. If f has a local maximum or minimum at x_0, then

$$\nabla f(x_0) = 0. \tag{6.11}$$

Proof. At a local extremum, the directional derivative must vanish along every direction v. Thus, by Equation 6.9,

$$D_v f(x_0) = \langle \nabla f(x_0), v \rangle = 0 \qquad \text{for every } v \in \mathbb{R}^n.$$

Since the inner product must vanish for every v, not just those perpendicular to the gradient, we must have $\nabla f(x_0) = 0$. □

Definition 6.1.8 (Critical point)**.** A point $x_0 \in U$ satisfying $\nabla f(x_0) = 0$ is called a *critical point* of f.

Since the gradient has n components, finding critical points amounts to solving n equations:

$$\frac{\partial f}{\partial x_1} = 0, \quad \frac{\partial f}{\partial x_2} = 0, \quad \ldots, \quad \frac{\partial f}{\partial x_n} = 0.$$

Example 6.1.9

Consider $f(x_1, x_2) = x_1^2 + 3x_1x_2$. Setting the gradient equal to zero, we get

$$\nabla f(x_1, x_2) = \begin{bmatrix} 2x_1 + 3x_2 \\ 3x_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies x_1 = x_2 = 0.$$

Thus, $(0, 0)$ is a critical point.

However, $\nabla f(x_0) = 0$ is only a necessary condition. A critical point may be a local minimum, a local maximum, or neither. To classify it, we need to examine higher-order derivatives.

§6.1.4 Multivariable Taylor expansion

Recall that in single-variable calculus, we used Taylor expansion to study extrema. We can extend the same idea to multivariable functions.

Let

$$x_0 = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \qquad h = \begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix}.$$

For a sufficiently smooth function, Taylor expansion around x_0 gives

$$f(x_0 + h) = f(x_0) + \frac{1}{1!}\langle h, \nabla f(x_0)\rangle + \frac{1}{2!}(h \cdot \nabla)^2 f(x_0) + \cdots, \tag{6.12}$$

where $h \cdot \nabla = \sum_{i=1}^{n} h_i \frac{\partial}{\partial x_i}$.

Now, suppose x_0 is a local minimum. Then $f(x_0 + h) - f(x_0) \geq 0$ for every sufficiently small h. But the first-order term $\langle h, \nabla f(x_0)\rangle$ changes sign when we replace h with $-h$. If this term were nonzero, it would dominate the higher-order terms for sufficiently small displacements, and the difference could not remain nonnegative in every direction.

Therefore, $\langle h, \nabla f(x_0)\rangle = 0$ for every h, which implies $\nabla f(x_0) = 0$. The same argument applies to a local maximum, where the difference must instead be nonpositive.

Thus, at a critical point, the first-order term vanishes. We now examine the second-order term to classify the critical point.

§6.1.5 The Hessian matrix

The second-order term in the Taylor expansion involves second-order partial derivatives, which we organise into a matrix called the Hessian.

Definition 6.1.10 (Hessian matrix)**.** Let $f : U \to \mathbb{R}$ be twice continuously differentiable. The *Hessian matrix* of f is the $n \times n$ matrix

$$H_f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}. \tag{6.13}$$

Since f is twice continuously differentiable, its mixed partial derivatives are equal, so the Hessian is symmetric, i.e., $H_f(x)^T = H_f(x)$.

The second-order term in Taylor's expansion can be written as

$$(h \cdot \nabla)^2 f(x_0) = h^T H_f(x_0) h = \langle h, H_f(x_0) h \rangle. \tag{6.14}$$

Here, $H_f(x_0)h$ is another vector in $\mathbb{R}^n$, and the expression is an inner product between this vector and h. The Hessian therefore describes the second-order behaviour of the function near x_0.

At a critical point, $\nabla f(x_0) = 0$, so Taylor's expansion becomes

$$f(x_0 + h) - f(x_0) = \frac{1}{2} h^T H_f(x_0) h + \cdots. \tag{6.15}$$

We can now use the sign of the second-order term to classify critical points.

§6.1.6 Classifying critical points

Recall that a symmetric matrix $A \in \mathbb{R}^{n \times n}$ is *positive definite* if $h^T A h > 0$ for every $h \neq 0$, and *negative definite* if $h^T A h < 0$ for every $h \neq 0$.

Proposition 6.1.11 (Second-order test for extrema)

Let $f : U \to \mathbb{R}$ be twice continuously differentiable, and let $x_0 \in U$ be a critical point. Then:

1. If $H_f(x_0)$ is positive definite, then x_0 is a strict local minimum.
2. If $H_f(x_0)$ is negative definite, then x_0 is a strict local maximum.

To see why, recall from Equation 6.15 that the leading second-order term is $\frac{1}{2} h^T H_f(x_0) h$. If the Hessian is positive definite, this term is positive for every nonzero h, and hence $f(x_0 + h) > f(x_0)$ for all sufficiently small nonzero h. Thus, x_0 is a strict local minimum. Similarly, a negative definite Hessian gives a strict local maximum.

Therefore, to find local extrema, we first solve $\nabla f(x_0) = 0$ and then evaluate $H_f(x_0)$ at the critical points. If the Hessian is neither positive nor negative definite, further examination may be necessary.

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